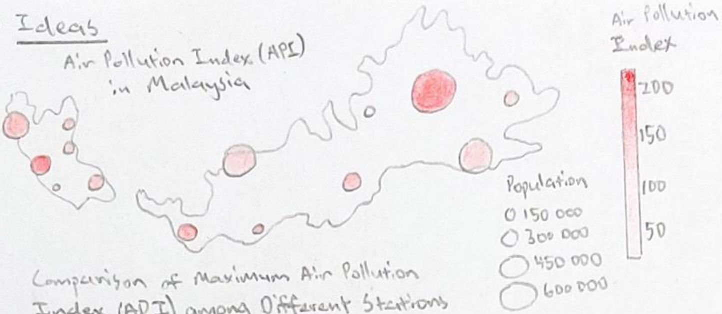
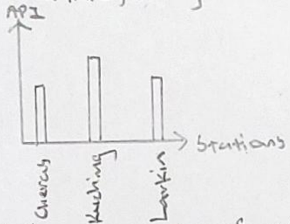
Sheet 1 Name: Tan Wei Liang Date: 19-9-2025

1. Ideas

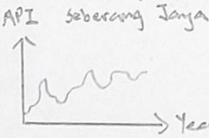
Air Pollution Index (API) in Malaysia



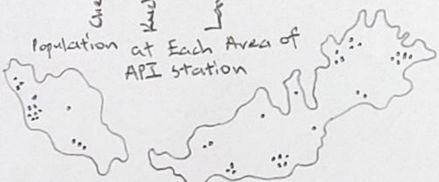
Comparison of Maximum Air Pollution Index (API) among Different Stations



Change in API across Different Years in Seberang Jaya

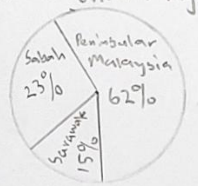


Population at Each Area of API station

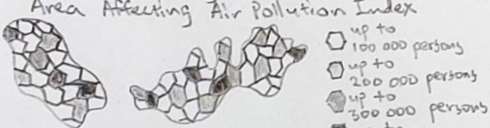


1 dot = 100 000 persons

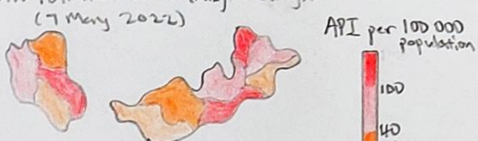
Percentage of Air Pollution Index Stations across Different Regions



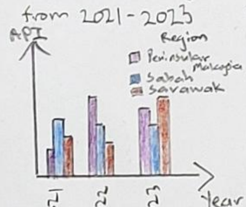
Population Bin Map Showing Density at Each Area Affecting Air Pollution Index



Air Pollution Index (API) in Malaysia (7 May 2022)

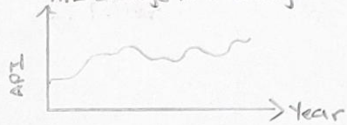


Air Pollution Index (API) of Different Regions from 2021-2023

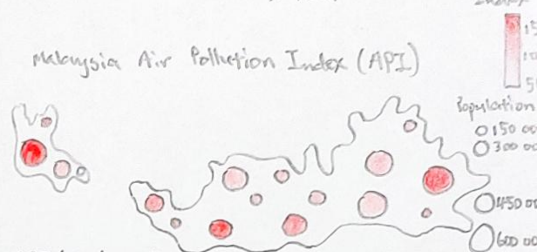


2. Filter

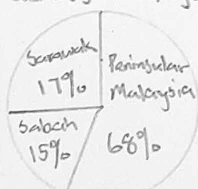
API Change in Kuching across Different Years



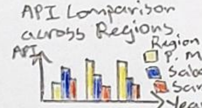
Malaysia Air Pollution Index (API)



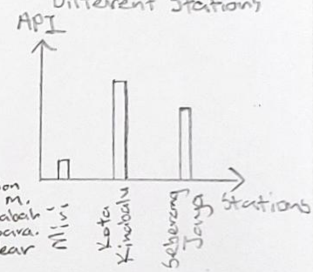
API Station Composition across Malaysia



API Comparison across Regions



Minimum Air Pollution Index (API) among Different Stations



3. Categorise

Horizontal/Vertical Bar chart → Air Pollution Index Comparison among different stations and areas

Proportional symbol map → Relationship between air pollution index and area population

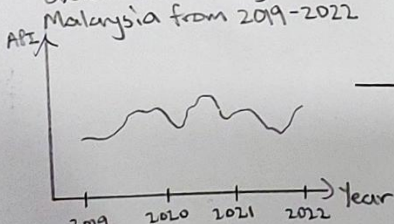
Pie chart → Analyse Station Count across Country

Line Chart → API trend analysis over time

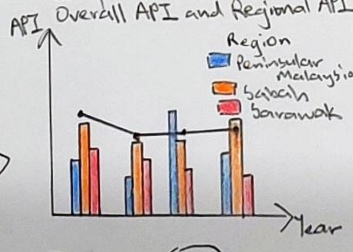
Grouped bar chart → API comparison between different regions across different years

4. Combine and Refine

Overall API Change in Malaysia from 2019-2022

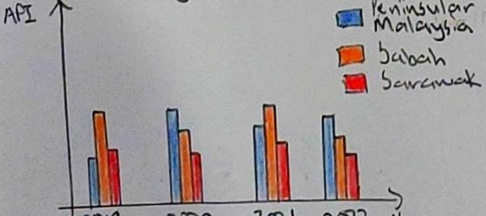


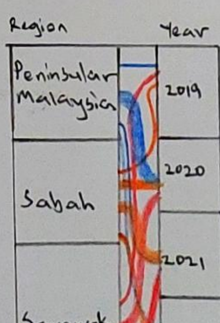
Comparison between Overall API and Regional API



or

API Comparison across Regions





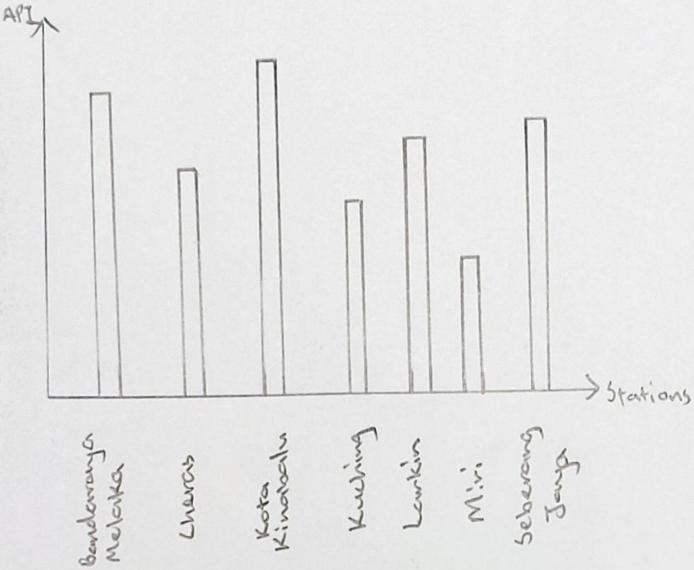
5. Questions

1) Is the pie chart needed?
2) Is the proportional symbol map suitable to be used?
3) Does the combined line-bar chart show important insight?

Sheet 2
Name: Tan Wei Liang
Date: 20-9-2025
Title: Initial ideas on Malaysia API
Description: Develop early visualisation concepts

Layout

Maximum API among Different Stations

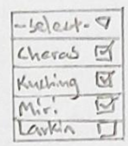


Air Pollution Index in Malaysia

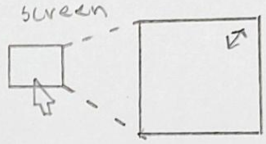
Operations

1. Dropdown menu to select stations to show API of stations on bar chart

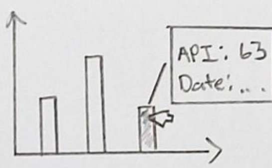
e.g.



2. Click on charts to access full screen



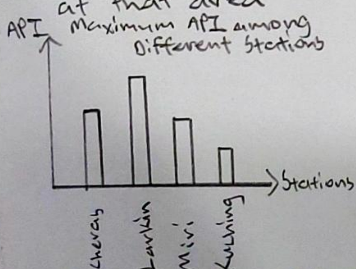
3. Hover for Details



Focus

Idea: Insights on Air Pollution Index

- Shows the difference in air pollution index between different selected stations to gain notion on the factors causing it
- Distinguish and compare the air pollution index which is affected by the population at that area



- Highlights station-wise variability, easy to identify hotspots, outliers
- encodes information through height of bars

Air Pollution Index in Malaysia

- Communicates spatial distribution of air pollution
- allows quick identification of regional disparities in air quality.

Discussion

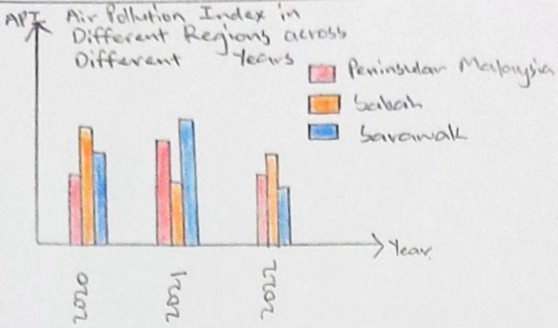
✓ Complementary perspectives
- Bar chart highlights detailed station-level differences
- choropleth map shows spatial distribution across Malaysia

✓ Intuitive interpretation
- familiar encodings used, easy to understand

X Potential for misinterpretation
- the choropleth map may suggest precise local variation even though pollution levels are measured at selected stations, which could mislead users for data accuracy

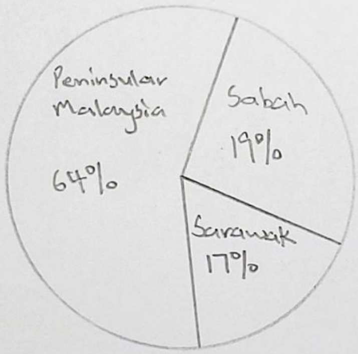
X Visual overload
- Presenting two different visualisation styles may overwhelm users

Layout



Area Population covering Each API Station

Percentage of Air Pollution Index Stations across Different Regions

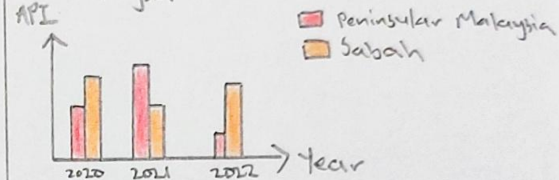


Sheet 3
Name: Tan Wei Liang
Date: 21-9-2025
Title: Dashboard Idea on Malaysia Air Pollution Index
Description: Design visualisation techniques to represent API trends in Malaysia

Operations

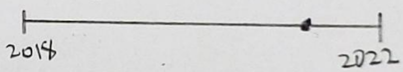
1. checkbox to filter region

☑ Peninsular Malaysia ☑ Sabah ☐ Sarawak

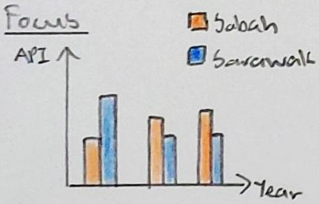


2. Highlight interaction, clicking on a region on pie chart will also highlight linked values or regions at other charts or maps

3. Time slider to select specific time range



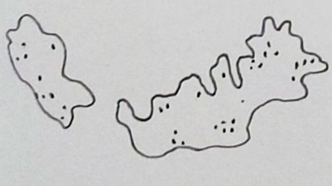
Focus



→ Display multiple regions across time, enabling direct comparison

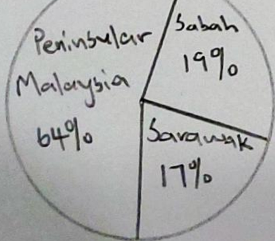
→ emphasizes regional trends over time

→ integrates both temporal and categorical dimensions



→ represents population coverage with dot density mapping

→ dot density method avoids distortion of region size



→ shows the share of API stations across Peninsular Malaysia, Sabah, Sarawak

→ Provides clear sense of proportion, highlighting regional imbalance in monitoring infrastructure

Discussion

✓ Clarity of representation

- each visualisation uses a simple, intuitive format (bars, dots, slices), making it easy for general audience to interpret without advanced statistical knowledge

✓ Multiple perspectives

- Provided holistic understanding by showing temporal change, population relevance, infrastructure distribution

X Fragmented insights

- Users need extra effort to mentally connect pollution levels, population exposure, station coverage

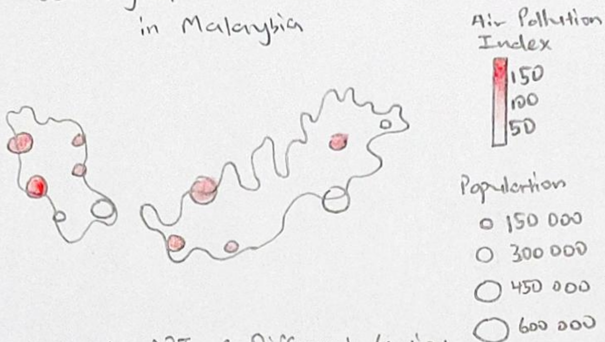
X Risk of oversimplification

- Dot map and pie chart mask finer details such as uneven pollution exposure within a region

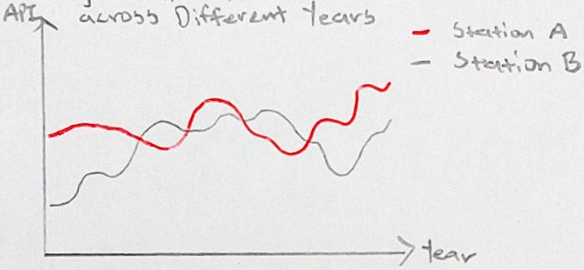
Layout

Sheet 4
Name: Tan Wei Liang
Date: 22-4-2025
Title: Screen idea on Malaysia Air Pollution Index
Description: Design interactive screens that explains Air Pollution Index in Malaysia

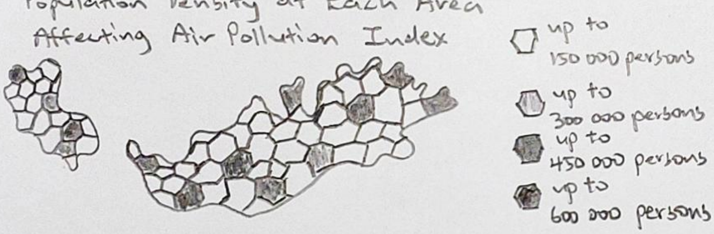
Severity of Air Pollution Index (API) in Malaysia



Change in API of Different Stations across Different Years

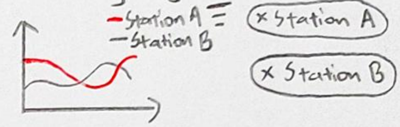


Population Density at Each Area Affecting Air Pollution Index

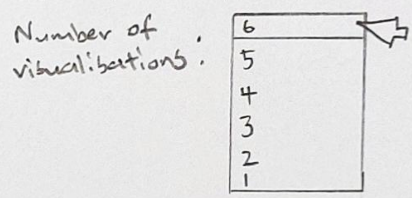


Operations

1. Filtering by Stations



2. Button to select number of visualisations to be shown



3. Slider to include less/more stations

Focus

① Proportional Symbol Map
- Combine geographical context with scaled circle symbols
- Size of each circle represents population density, fill color encodes Air Pollution Index (API)
- Its focus lies in dual encoding: size captures demographic influence (population) and color captures pollution severity

② Line chart
- Show temporal variation in API at different stations

 } Focus: Comparability of multiple stations over time

③ Bin Map
- Uses spatial binning technique (polygons)
- Focus: aggregating complex population data into uniform spatial units

Discussion

✓ Multi-faceted insights
- Captures spatial, temporal, demographic dimensions of air pollution

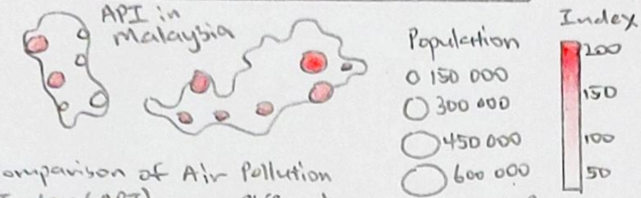
✓ Effective pattern recognition
- Easy for users to spot correlations and outliers

✗ Cognitive load
- Multiple visual encodings may overwhelm users

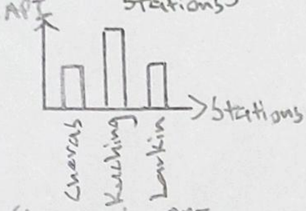
✗ Potential misinterpretation
- Symbol size and color intensity in the map can be hard to compare accurately

Layout

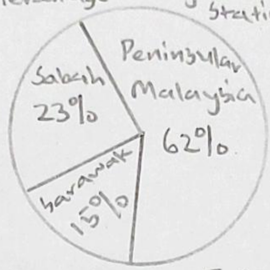
An Overview of Air Pollution Index in Malaysia



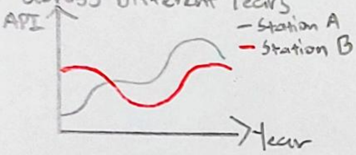
Comparison of Air Pollution Index (API) among Different Stations



Percentage of Regional API Stations



Change in API across Different Years

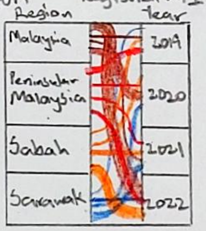


Air Pollution Index (API) of Different Regions from 2020-2022

Standardised Air Pollution Index (API) in Malaysia (6 May 2022)

API per 100 000 population

Overall API vs Regional API

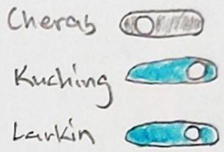


Sheet 5
Name: Tan Wei Liang
Date: 23-9-2025
Title: Final API Implementation Design
Description: Visualisation Idiom Decisions

Operations

1. API stations toggle to turn on/off station displays on charts

e.g.



2. Sorting (Highest to Lowest API or vice versa)

Sort : Ascending | Descending

3. Click on the topic to have full description about Malaysia Air Pollution Index

An Overview of Air Pollution Index in Malaysia

Focus

① Proportional Symbol Map
- circles represent population size
- color intensity shows API levels
- Focus: dual encoding (size + color), reveals both demographic concentration and pollution severity

② Bar chart
- Compare API levels across monitoring stations
- Focus: Direct comparability: can easily spot stations with highest/lowest API

③ Pie chart
- shows proportion of monitoring stations across Malaysia
- Focus: Contextualizing station distribution

④ Line chart → Display temporal trends for stations
Focus: Link spatial and temporal perspectives

⑤ Grouped bar chart → Compare API values of Different Regions
Focus: Showing temporal progression by region

⑥ Choropleth Map → Uses continuous color shading
Focus: Standardization by population

Details

- Statistical Standardisation
  - API values standardized per 100 000 population would be:

$$API_{Standardized} = \frac{API}{Population} \times 100\,000$$

- Trend analysis: Line charts and grouped bar charts rely on time-series aggregation

- Proportional calculations: Pie chart percentages are derived using:

$$\% = \frac{\text{Stations in Region}}{\text{Total Stations}} \times 100$$

- Software libraries: Vega-Lite for web-based, interactive visualisations

- Compatibility: Designed to integrate with existing platforms like web dashboards.